Biokinetic Valorization of Camu Camu Peel (Myrciaria dubia) as a Renewable Energy Source for the Ucayali Region

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INTRODUCTION

Camu camu (Myrciaria dubia) is an Amazonian fruit of agro-industrial importance. Its processing generates waste, mainly peels.

These residues contain organic matter and nutrients that can be harnessed through biological processes. Anaerobic digestion transforms organic matter into biogas, a renewable energy resource.

HIGHLIGHTS OF THE STUDY

- Anaerobic digestion treats organic waste while recovering renewable energy.
- Biogas production depends on substrate composition and variables such as pH, temperature, C/N ratio, and retention time.
- Valorizing agricultural waste contributes to the circular economy and resource recovery.

PROBLEM CONTEXT

In 2025, direct exports of Peruvian camu camu reached approximately $1.74 million USD , reflecting its agro-industrial potential (PROMPERÚ, 2026). However, nearly 42% of the fruit's weight consists of by-products, including peel and seeds (scientific review, 2023). Anaerobic digestion could convert these residues into biogas, helping reduce improper disposal, promote the circular economy, and explore clean-energy alternatives for Amazonian communities in Ucayali.

REFERENCE LOCATION



Department: Ucayali
Province: Coronel Portillo
Districts: Callería, Yarinacocha, Manantay

Region of biomass origin

Map source: IDE Ucayali / Regional Government of Ucayali.

METHODOLOGY

Experimental research • quantitative approach

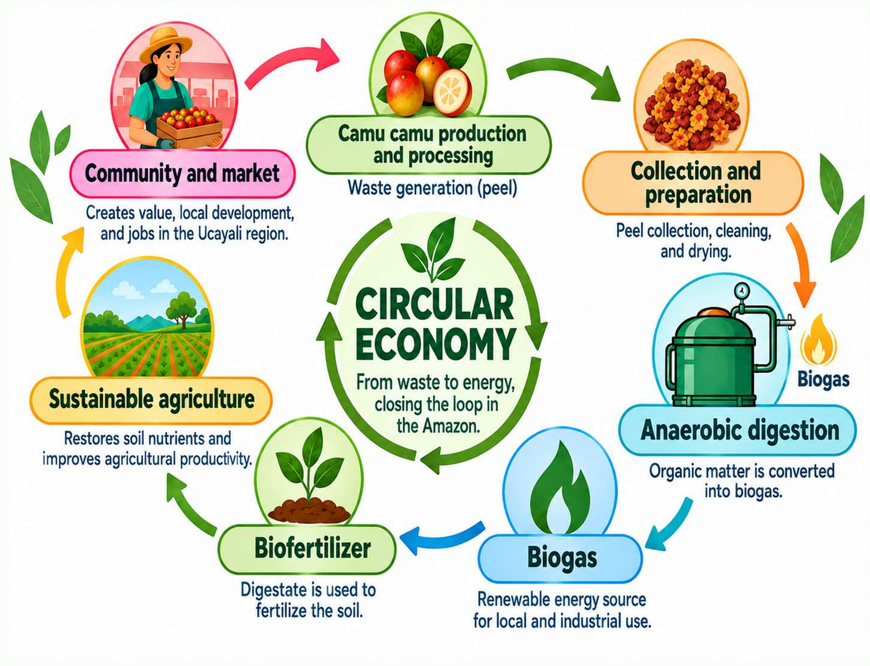
TREATMENTS / EXPERIMENTAL DESIGN

T0	Control
T1	Camu camu peel (25%)
T2	Camu camu peel (50%)

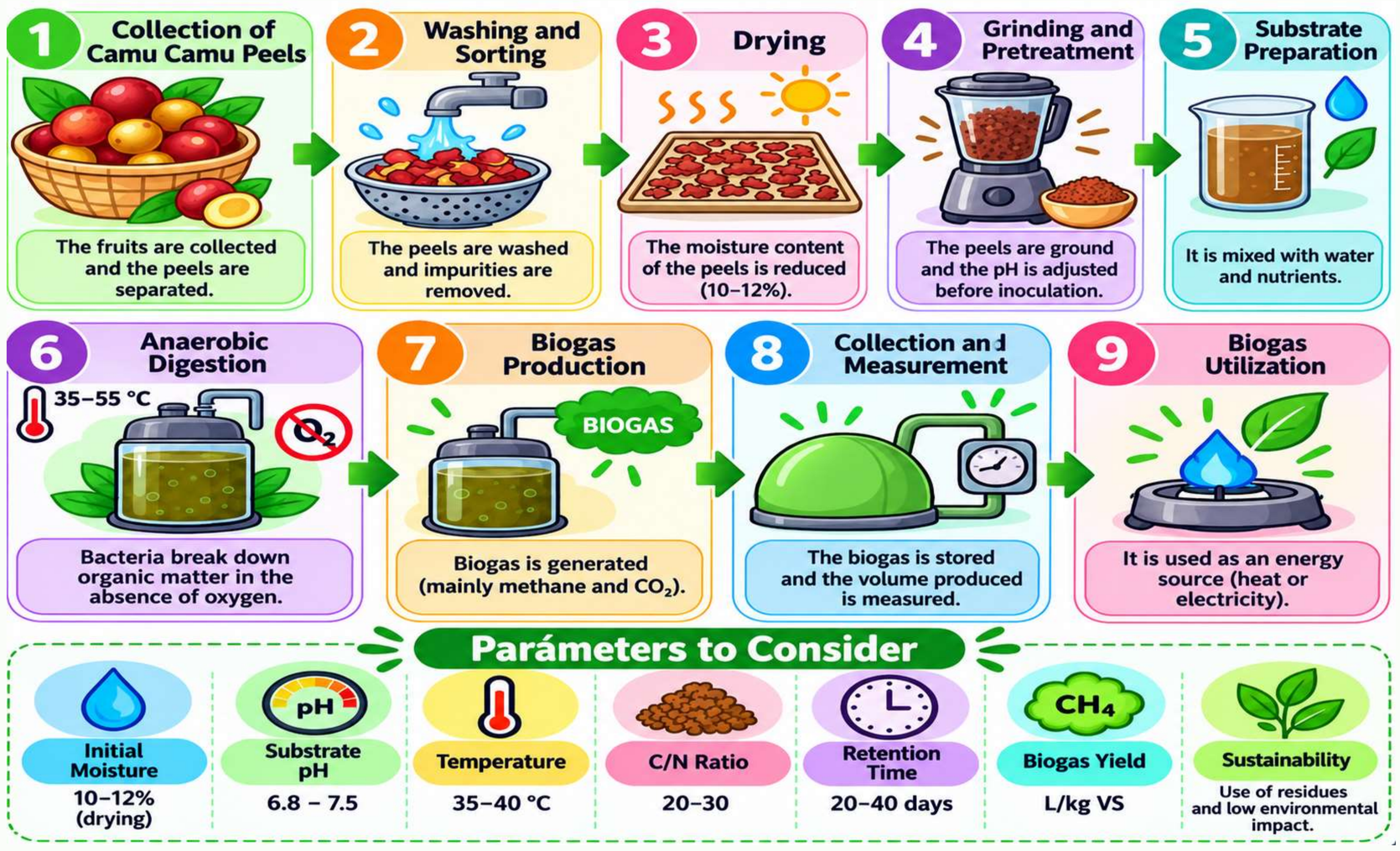
PARAMETERS TO BE EVALUATED

Biogas production	Volume produced
Yield	Biogas / substrate
Production time	Generation period
Substrate characterization	Physicochemical parameters

VALORIZATION AND CIRCULAR ECONOMY



EXPERIMENTAL PROCESS SEQUENCE



EXPECTED RESULTS

This study is expected to determine the potential of camu camu peel as a substrate for biogas production by assessing gas volume, yield, and generation time.
The findings are expected to provide evidence of the feasibility of recovering energy from this agro-industrial residue, helping explore an alternative that is more environmentally friendly and beneficial to public health.



TECHNOLOGICAL IMPLICATIONS

ENVIRONMENTAL	ENERGY	AGRO-INDUSTRIAL	SCIENTIFIC
Waste recovery and reduction of improper disposal.	Potential biogas production from residual biomass.	Valorization of a by-product from camu camu processing.	Generating knowledge about the energy potential of an Amazonian residue.

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